

CHRIS

AI-Based Autonomous Hazard Detection and Exploration Robot Using Orientation-Aware Adaptive Electromagnetic Scanning

Final Presentation

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Problem Statement

Invisible Electrical Hazards

Live conductors, overloaded circuits, and leaking electromagnetic fields are often imperceptible to human senses until a fatal incident occurs.

Static Sensors Insufficient

Fixed alarm systems lack mobility. They cannot follow a hazard as it moves, nor can they correlate EM field direction with physical source location.

Manual Inspection Risk

Electrical inspection in live environments exposes technicians to arc flash, shock hazard, and toxic gas. High-risk zones demand remote or robotic monitoring.

Need: Intelligent Mobile System

A robot that autonomously patrols, detects multi-modal hazards, estimates EM source direction, interacts via voice, and generates persistent hazard maps.

Objectives — All Achieved ✓

✓ Detect humans & obstacles using camera YOLO + HC-SR04 ultrasonic sensor

✓ Identify electrical hazards via proposed EMI coil + MPU-6500 orientation-aware scanning

✓ Provide real-time audio alerts and voice interaction using Groq LLaMA 70B AI + TTS

✓ Implement autonomous indoor patrol with obstacle avoidance and 360° scanning

✓ Integrate cloud AI for advanced hazard analysis, voice commands and intelligent responses

✓ Build orientation-aware EM field directional mapping — $\theta^* = \operatorname{argmax} E(\theta, \varphi)$

Literature Review

Author / Source	Year	Publication	Title	Key Contribution
Khamis et al.	2026	Scientific Reports	RoboFusion: Hybrid real-synthetic dataset for robotic hazard detection in industrial AMR systems	Multi-sensor AMR with 12-sensor suite; 180-day industrial deployment; IoT-cloud hazard mapping
Simamora et al.	2025	arXiv	Indoor Air Quality Detection Robot using IoT with MQ-2, DHT11, MPU6050, VL53L0X	ESP32 mobile robot with gas, temp/humidity, IMU, distance sensors; Mamdani fuzzy classifier
Sotomi et al.	2025	arXiv (NYU)	Embodied Hazard Mitigation using Vision-Language Models for Autonomous Mobile Robots	VLM + LLM for anomaly detection; 91.2% accuracy; edge-AI hazard mitigation actions
Muñoz et al.	2019	MDPI Sensors	Gas sensor array in assistant robot for early gas leak detection	MOX sensor array on mobile robot; indoor odour mapping; hazard zone identification
IJET	2024	Int. J. Eng. Tech.	AI-Enabled IoT Autonomous Air Quality Monitoring on Wheeled Robot Platform	Arduino robot; CO ₂ + temperature; AI multi-level warning (W1–W4); autonomous shutdown at critical gas
IEEE	2014	IEEE Xplore	Autonomous mobile robot for indoor security patrol using sensor fusion	Wheeled security robot; IMU + ultrasonic sensor fusion; intelligent autonomous patrol control

Full Robot Block Diagram

CHRIS Robot — Full Hardware Block Diagram

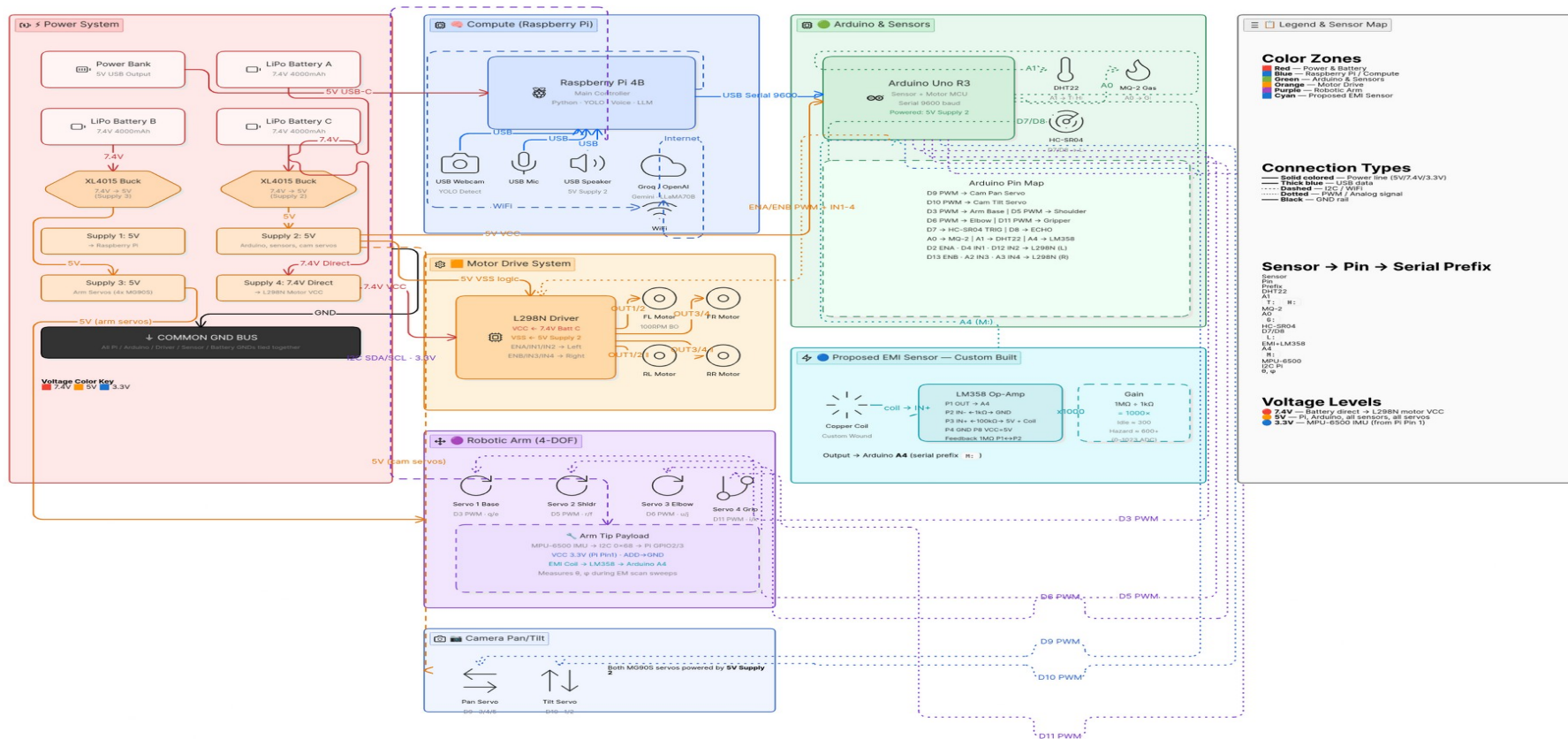


Fig. 0 — CHRIS Robot Full Hardware Block Diagram

Software Architecture

Main loop · Patrol loop · Arm EM scan · AI integration

main.py — Software Module Map (~7000 lines, single file)

Voice Input

Google Speech API
WebRTC VAD
listen_and_transcribe()

AI Brain — Groq LLaMA 70B

send_text_to_gpt4_move()
physical + speech + mental cmds
Cloud API via WiFi

Camera + YOLO

OpenCV USB webcam
YOLO object detection
yolo_detect()

Movement Control

w/s/a/d/x → Arduino
Patrol loop
Obstacle avoidance

EIL Innovation Module

MPU-6500 orientation θ, φ
Adaptive scan $P=f(\hat{E}, \nabla E, U, d\hat{E}/dt)$
_patrol_arm_em_scan()

Sensor Reading

Gas/Temp/Humidity/EMI
Prefixed serial G: T: H: M:
_serial_read_prefixed()

4-DOF Arm Control

move_arm_to()
arm_em_scan()
tip_direction_scan() i/k

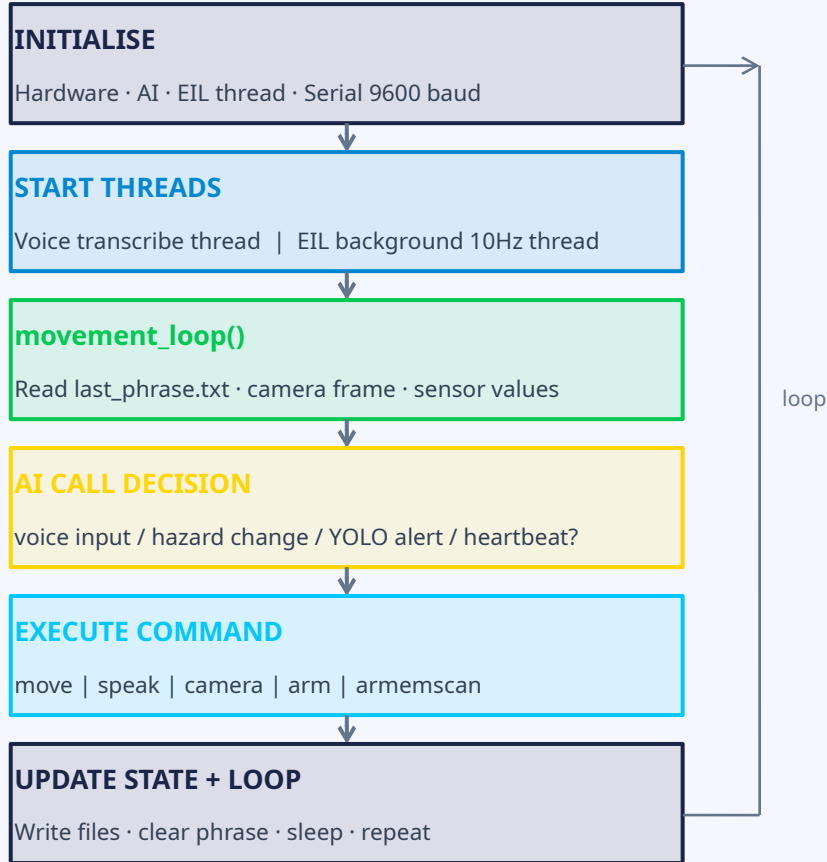
Hazard & Mapping

Threshold checks
Zone memory JSON
Heatmap HTML generation

Speech Output

pyttsx3 TTS
playback_text.txt
60s dedup cooldown

Main Loop Flowchart



Parallel Threads:

- Voice Thread: continuous listen → Google STT → last_phrase.txt
- EIL Thread: 10Hz → read MPU-6500 → update $E(\theta, \varphi)$ grid → eil_status.json
- Image Thread: capture frame → YOLO → annotated output1.jpg → WebSocket upload

AI Call Triggers:

- New voice input (last_phrase $\neq$ *No Mic Input*)
- Hazard state change detected
- YOLO detects person in frame
- Patrol heartbeat — every 3 patrol spots
- Interactive heartbeat — every 5 min idle

Output Files:

- last_phrase.txt · playback_text.txt · eil_status.json
- hazard_zone_memory.json · emi_latest.txt · output1.jpg

Patrol Loop Flowchart

PHASE 1: MOVE FORWARD

reliable_distance(3 samples) → discard 200cm timeouts → median
>25cm: forward 2s | Blocked: avoid_obstacle()

PHASE 2: 360° SCAN — 8 × 45°

Turn 45° → stop → camera sweep → read all sensors → zone memory update
Camera: 3=pan-left 4=center 5=pan-right 2=tilt-up 1=tilt-center

PHASE 3: ARM EM SCAN (every spot)

_patrol_arm_em_scan() | Adaptive COARSE/FOCUSED/DENSE from P score
Speech: "Starting EM scan..." → "Scan complete at spot N"

PHASE 4: AI HEARTBEAT (every 3 spots)

Sensor summary injected → Groq AI assesses → may speak/move/alert
MAX_PATROL_SPOTS = 30 | REPEAT

Obstacle Avoidance:

- Try right 90° → check distance
- If clear → continue forward
- Else try left 180° → check
- If still blocked → reverse 1.2s

Zone Memory (hazard_zone_memory.json):

- Key: "spot1_sector2" — unique per position + direction
- priority = hazard_count / visit_count
- High-priority sectors scanned first next patrol
- Persists across power cycles — robot learns environment

Serial Commands (Arduino):

- w=forward s=backward a=left d=right x=stop
- l=distance g=gas t=temp h=humidity m=EMI
- q/e=arm-base r/f=shoulder u/j=elbow i/k=gripper

Arm EM Scan — Software Flowchart



Adaptive Pose Count:

COARSE $P \leq 0.35$

3 poses ~3s no tip

FOCUSED $P \leq 0.65$

12 poses ~12s no tip

DENSE $P > 0.65$

18 poses +i/k tip sweep ~25s

Outputs per scan:

- Peak pose: θ^* and φ^* — EM source direction
- Peak EMI value E^* — field strength
- arm_em_scan_results list → arm_em_scan_DATE.json
- eil_status.json updated with tip_theta_star, tip_E_hat
- Speech: "Arm scan complete. Peak field at tilt N degrees"

i/k Tip Sweep (DENSE):

angles = $[-30^\circ, -15^\circ, 0^\circ, +15^\circ, +30^\circ]$ → $\theta^*_{tip} = \text{argmax } E$ over 5 readings
move_tip(steps) sends i or k commands to Arduino servo

```
[PATROL] + Patent arm EM scan at spot 1...
[ARM EIL SCAN] Adaptive - mode=FOCUSED P=0.423
[ARM EIL SCAN] FOCUSED: 12 poses - homing in
[F] base= 45° sh= 70° el= 70° | EMI=142.0 | θ=-12.3°
[F] base= 45° sh= 70° el=110° | EMI=187.0 | θ=-34.7°
[F] base= 45° sh=110° el= 70° | EMI=203.0 | θ=-19.2°
[F] base= 90° sh= 70° el= 70° | EMI=241.0 | θ= -8.1° -PEAK
[F] base= 90° sh= 70° el=110° | EMI=198.0 | θ=-15.4°
[F] base=135° sh= 70° el= 70° | EMI=112.0 | θ=+15.2°
[ARM EIL SCAN] Peak EMI=241.0 at base=90° sh=70°
[ARM EIL SCAN] θ*=-8.1° φ*=5.0° → EM source direction
Arm scan complete at spot 1. Peak at tilt -8 degrees.
```

[illegible][illegible]

```

Image sent successfully in 0.003 seconds.
false
false
false
false
true -- Consolidated rough draft memory from past sessions.
name flag: false
name text: false
Image sent successfully in 0.003 seconds.

=====
[PATROL] spot=1 sector=1/8 zone=spot1_sector0 cam=4
Temp : 30.6°C ok
Humidity : 76.0% ok
Gas : 245 ok
EM coil : 566.0 ok Distance: 200cm
EIL zone : WARNING P=0.554 FOCUS_SCAN
M_t grid : 0.9842 zone_visits=11
=====
Image sent successfully in 0.003 seconds.
false
false
false
false
true -- Consolidated rough draft memory from past sessions.
name flag: false
name text: false
Image sent successfully in 0.003 seconds.
[AUDIO] Google: could not understand audio

```

[illegible][illegible]

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Software — Hazard Heatmaps (HTML Browser Output)

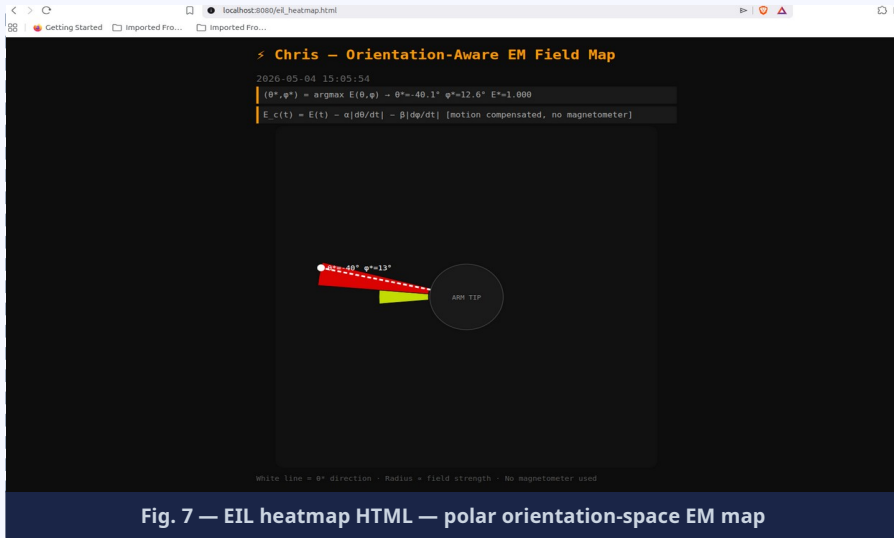


Fig. 7 — EIL heatmap HTML — polar orientation-space EM map

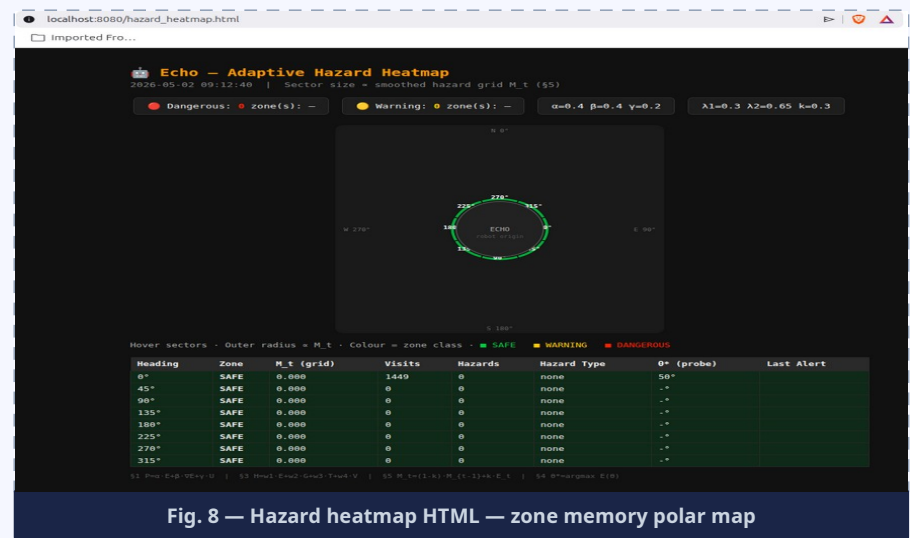
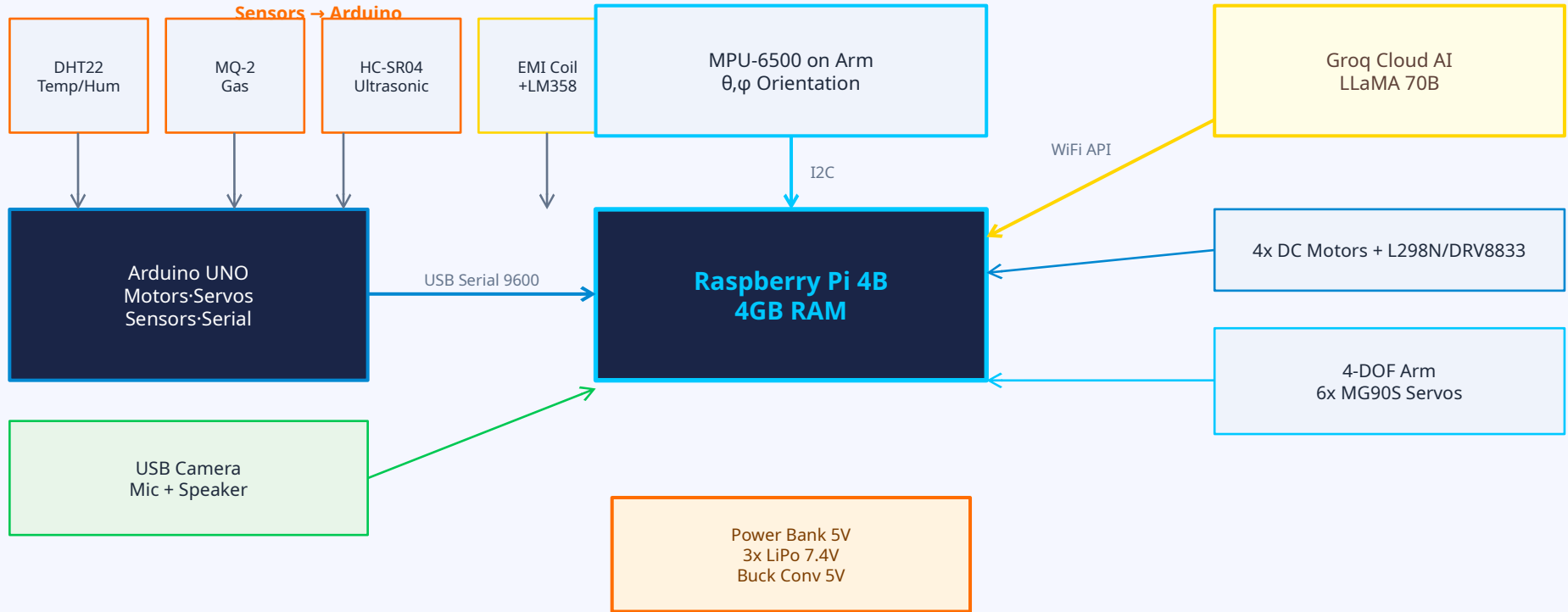


Fig. 8 — Hazard heatmap HTML — zone memory polar map

Hardware Implementation

Components · Connections · Power Architecture

Hardware Architecture Overview



All subsystems share common GND bus | Logic 5V rail isolated from motor 7.4V rail

Component Specifications

Component	Model / Spec	Key Parameters	Function in CHRIS
Raspberry Pi 4B	4GB RAM, 1.5GHz Quad-core	WiFi, USB3, 40-pin GPIO	Main AI + vision + EIL controller
Arduino Uno R3	ATmega328P, 16MHz	14 digital, 6 analog I/O	Firmware, motors, servos, serial
MPU-6500 (on arm)	6-axis IMU — I2C 0x68	$\pm 250^\circ/\text{s}$ gyro, $\pm 2\text{g}$ accel	Arm orientation θ, φ tracking
Proposed EMI Coil+LM358	DIY induction coil + LM358	Gain = 1000 $\times$, A4 output 0-1023	EM field strength sensing
DHT22	Temp + Humidity	$\pm 0.5^\circ\text{C}$, $\pm 2\%$ RH	Environmental hazard monitoring
MQ-2 Gas Sensor	Analog 5V	ADC 0-1023, LPG/CH ₄ /Smoke	Gas and smoke detection
HC-SR04 Ultrasonic	2–400 cm, $\pm 3\text{mm}$	5V, TRIG/ECHO digital pins	Obstacle detection (fixed front)
USB Webcam	1080p30	USB, YOLO object detection	People, fire, hazard vision
L298N/drv8833 Motor Driver	7–12V, 2A/ch	ENA/ENB PWM, IN1–IN4	Controls 4 DC drive motors
MG90S Servo $\times 6$	4.8–6V metal gear	180 $^\circ$, 2.0 kg $\cdot\text{cm}$	Arm joints + camera pan/tilt
LiPo Battery 7.4V $\times 3$	4000mAh, 2S LiPo	$\sim 7.4\text{V}$ nominal, 20C discharge	Power for motors, servos, sensors
Power Bank (Pi supply)	5V USB output	2–3A output	Stable 5V to Raspberry Pi
XL4015 Buck Conv. $\times 2$	7.4V $\rightarrow$ 5V, 5A	Adjustable output, efficient	Regulated 5V for servos + sensors

Drive System & Power Architecture

Drive System

- 4× BO Geared DC Motors — 100 RPM, dual shaft
- L298N H-Bridge motor driver (7–12V, 2A/ch)/DRV8833
- PWM speed control via ENA/ENB Arduino pins
- Battery C (7.4V LiPo) → L298N(70%)/DRV8833(95%) DIRECT — no buck
- Commands: w=fwd s=back a=left d=right x=stop
- Turn timing: $45^\circ \approx 0.55s$ $180^\circ \approx 1.2s$
- Forward per step: 2s (~30cm at medium speed)
- Obstacle: right→left→back avoidance sequence
- Patrol: forward first → 360° scan → arm EM scan

Power Architecture

Supply 1

Power Bank 5V USB → Raspberry Pi (direct, stable)

Supply 2

LiPo A 7.4V → XL4015 Buck → 5V: Camera servos + DHT22 + MQ-2 + HC-SR04 + Speaker

Supply 3

LiPo B 7.4V → XL4015 Buck → 5V: Arm servos (all 4 × MG90S)

Supply 4

LiPo C 7.4V → L298N/DRV8833 DIRECT (no buck): 4× DC drive motors

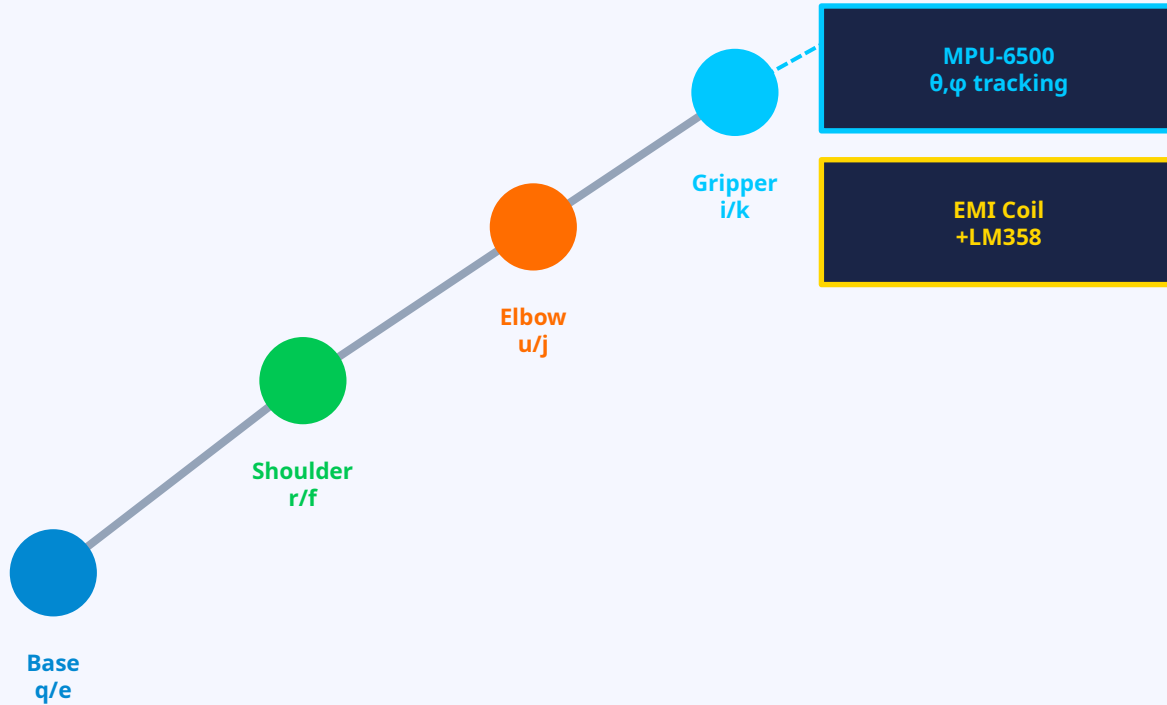
GND

Common GND bus — all supplies, Pi, Arduino, sensors, driver

Note

Physical isolation: motor 7.4V rail $\neq$ logic 5V rail → stable operation

4-DOF Robotic Arm — Hardware



Servo motors:

6× MG90S metal gear, 180°, 2.0 kg·cm, 4.8–6V

Joints:

Base · Shoulder · Elbow · Gripper (4 DOF)

Power:

Supply 3: LiPo B 7.4V → XL4015 → 5V dedicated rail

MPU-6500:

Mounted at arm tip, I2C to Pi (SDA GPIO2, SCL GPIO3)

EMI coil:

Custom coil at arm tip → LM358 → Arduino A4

ARM_STEP:

10° per i/k command, limits 20°–160°

Home pose:

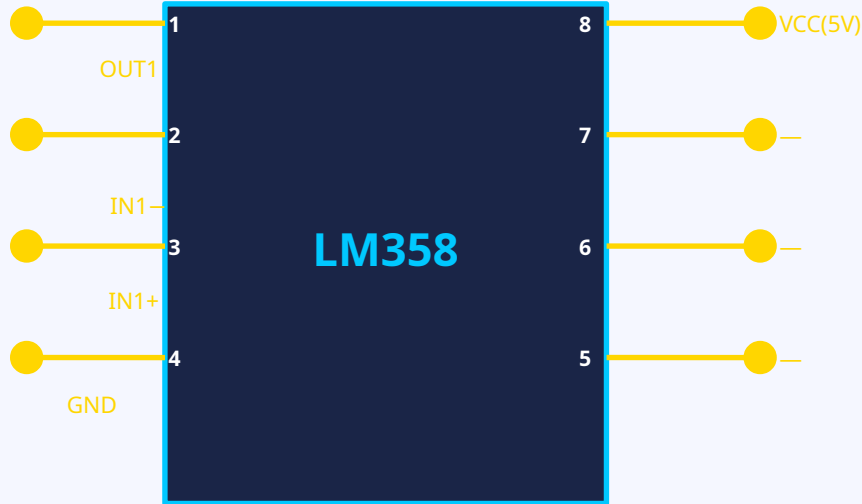
base=90 sh=90 el=90 grip=30

Scan poses:

COARSE:3 FOCUSED:12 DENSE:18+tip sweep

Proposed EMI Sensor — Custom Built Amplifier Circuit

LM358 Custom Circuit



Final Wiring

From	To (LM358)	Component	Purpose
5V supply	Pin 3 (IN1+)	100kΩ (R ₁)	Upper bias resistor
Pin 3 (IN1+)	GND	100kΩ (R ₂)	V _{mid} = 2.5V (lower bias)
Coil End 1	Pin 3 (IN1+)	Direct wire	EMI signal on 2.5V midpoint
Coil End 2	GND	Direct wire	Coil reference/return
GND	Pin 2 (IN1-)	1kΩ (R _{in})	Gain input resistor
Pin 1 (OUT1)	Pin 2 (IN1-)	1MΩ (R _f)	Feedback → gain=1000×
Pin 1 (OUT1)	Arduino A4	Direct wire	Output to ADC 0–1023
5V supply	Pin 8 (VCC)	Direct wire	Op-amp power (+5V)
GND	Pin 4 (GND)	Direct wire	Op-amp ground ref
Pin 8	Pin 4	0.1μF ceramic	Noise suppression (IMPORTANT)

Gain = $1\text{M}\Omega \div 1\text{k}\Omega = 1000\times$ | Coil → AC EM field → μV → LM358 amplifies → Arduino A4 reads 0–1023 | Prefixed "m" command → M: response

Calibration:

Idle ~200-212 | Near 12V adapter ~400+ | Threshold: deviation >200 = hazard

How it works:

100kΩ bias stabilises Pin 3 → coil detects AC field → LM358 amplifies $\times 1000$ → Pi reads via Arduino "m" command

Hardware — Construction & Assembly

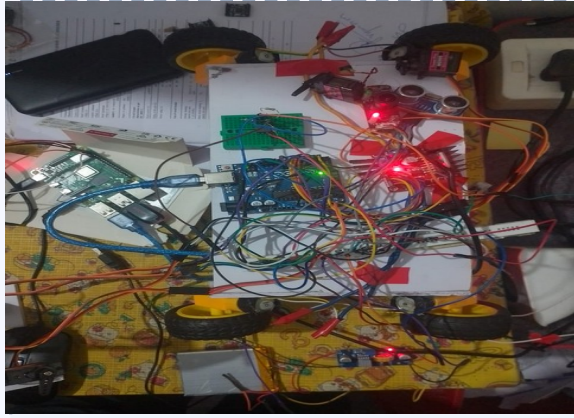


Fig. 9 — Top view of robot during construction



Fig. 10 — Side view of robot during construction



Fig. 11 — Full assembled robot overview

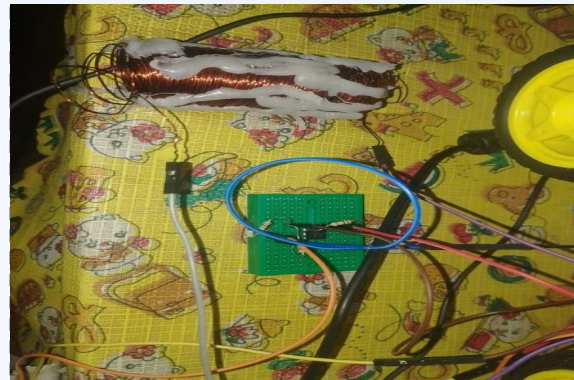


Fig. 12 — EMI coil + LM358 IC connected

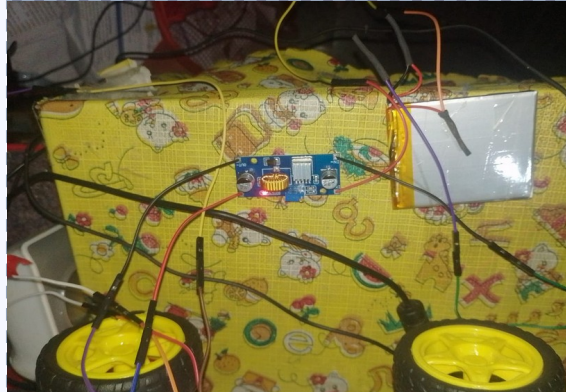


Fig. 13 — Buck converter connected to LiPo battery

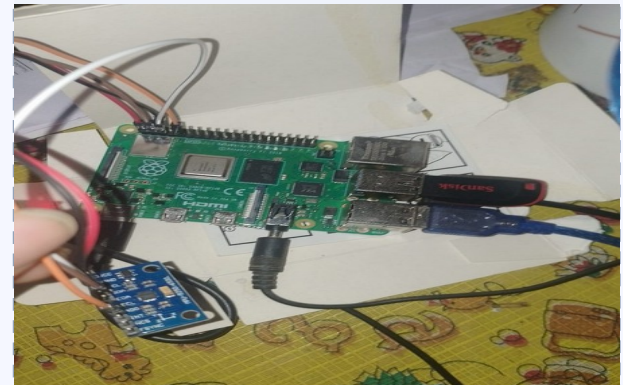


Fig. 14 — Raspberry Pi connected to MPU-6500

Hardware — Testing & Measurements

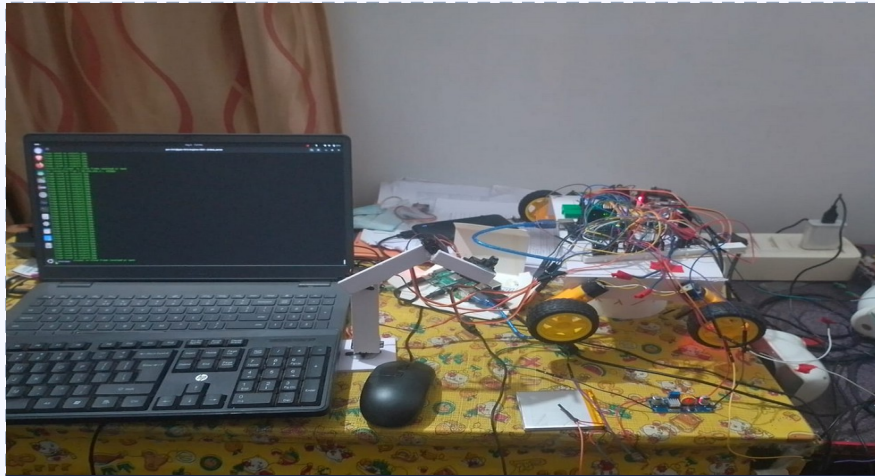


Fig. 15 — Full robot with laptop — program running in terminal

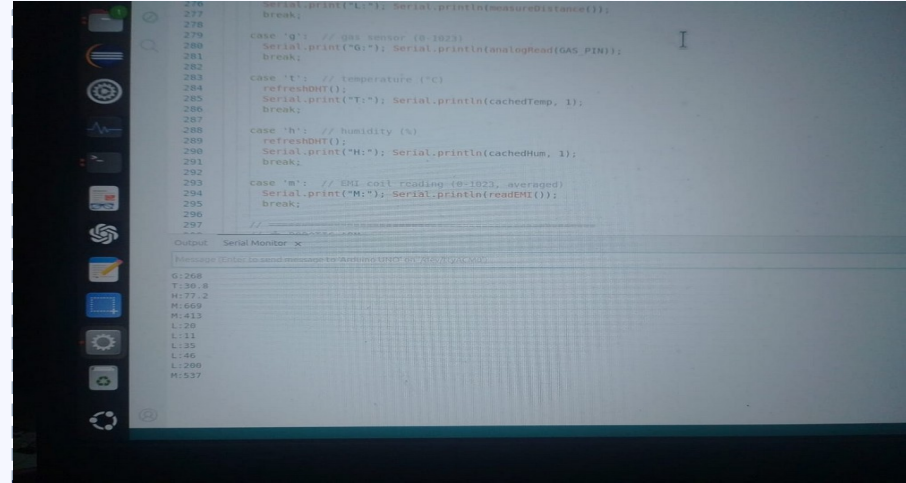


Fig. 16 — Arduino IDE: serial commands and sensor responses

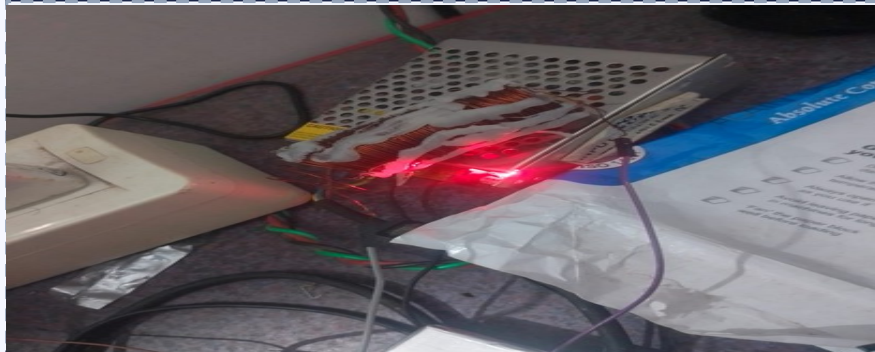


Fig. 17 — EMI coil near 12V adapter — EM field measurement

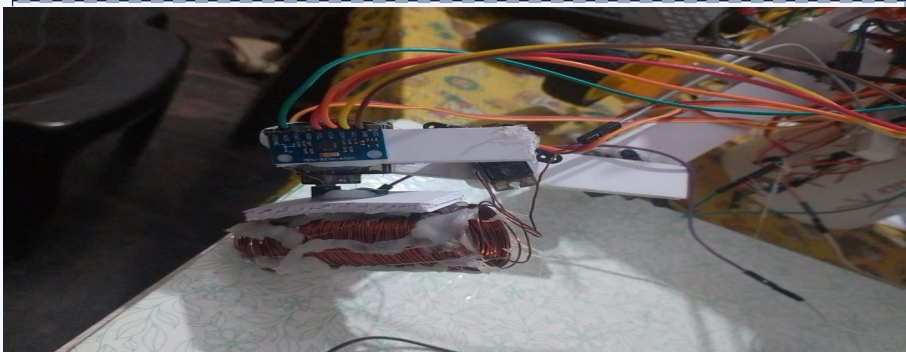


Fig. 18 — MPU-6500 mounted on robotic arm (close-up)

Core Innovation

Adaptive EM Field Mapping · Equations · Direction Estimation

Orientation-Aware EM Mapping — Core Concept

✗ Conventional Approach

Magnetometer reads one direction only
at a fixed body position

$E(t)$ = single scalar value

No spatial direction information
Cannot locate EM source bearing

Dependent on expensive compass chip
Susceptible to magnetic interference

✓ CHRIS Innovation

Arm sweeps 18 poses across 3D space
MPU-6500 on arm: real θ, φ at each pose
EMI coil reads E at each orientation

Builds map: $E(\theta, \varphi)$

$\theta^* = \operatorname{argmax} E(\theta, \varphi)$
→ Direction of maximum flux linkage
→ EM source bearing WITHOUT compass

Adaptive: more poses when signal rises

Mathematical Equations — Orientation & Compensation

Eq 1 — Complementary Filter (Orientation Estimation)

$$\begin{aligned}\theta_t &= \alpha \cdot (\theta_{t-1} + \omega_x \cdot \Delta t) + (1-\alpha) \cdot \arctan(a_x / \sqrt{a_y^2 + a_z^2}) \\ \varphi_t &= \alpha \cdot (\varphi_{t-1} + \omega_y \cdot \Delta t) + (1-\alpha) \cdot \arctan(a_y / a_z)\end{aligned}$$

$\alpha = 0.96 \rightarrow$ gyro trusted 96%, corrected by accelerometer angles | Stable pitch θ and roll φ — no magnetometer required

Eq 2 — Motion Compensation (removes arm-motion noise)

$$E_c(t) = E(t) - \alpha_m \cdot |d\theta/dt| - \beta_m \cdot |d\varphi/dt| \quad \alpha_m = 0.30, \beta_m = 0.20$$

Subtracts noise induced by arm swinging through ambient field | Fast arm movement $\rightarrow$ large $d\theta/dt \rightarrow E_c$ corrected downward

Eq 3 — Running Normalisation

$$\begin{aligned}\hat{E} &= (E_c - E_{\min}) / (E_{\max} - E_{\min} + \varepsilon) \\ \varepsilon &= 1 \times 10^{-6}\end{aligned}$$

Dynamic range — no static pre-calibration needed

Eq 4 — Uncertainty Model

$$\begin{aligned}U(\theta, \varphi) &= 1 / (1 + N_{\text{visits}}(\theta, \varphi)) \\ U &\in (0, 1], \quad U=1 \text{ at first visit}\end{aligned}$$

Encourages exploration of unvisited orientations

$\alpha=0.96$: 96% gyro + 4% accel correction each step | $\beta_m=0.20$ removes roll-induced noise | Motion comp. is essential during DENSE tip sweep

Mathematical Equations — Priority, Grid & Decision

Eq 5 — Scan Priority Score P

$$P = w_1 \cdot \hat{E} + w_2 \cdot \nabla E + w_3 \cdot U + w_4 \cdot \max(d\hat{E}/dt, 0) \quad w_1=0.35 \quad w_2=0.30 \quad w_3=0.20 \quad w_4=0.15$$

Computed every 100ms by EIL thread | $P \leq 0.35 \rightarrow \text{COARSE}(3 \text{ poses})$ $P \leq 0.65 \rightarrow \text{FOCUSED}(12 \text{ poses})$ $P > 0.65 \rightarrow \text{DENSE}(18 + \text{tip sweep})$

Eq 6 — Optimal Direction

$$(\theta^*, \varphi^*) = \operatorname{argmax}_{\{\theta, \varphi\}} E(\theta, \varphi)$$

Peak EM source bearing from orientation map

Eq 7 — Tip Sweep (DENSE mode)

$$\theta^*_{\text{tip}} = \operatorname{argmax}_{\{\theta \in \{-30^\circ, -15^\circ, 0^\circ, +15^\circ, +30^\circ\}\}} E(\theta)$$

i/k gripper commands — sub-degree precision

Eq 8 — Hazard Grid Update

$$M_t(\theta, \varphi) = (1-k) \cdot M_{\{t-1\}}(\theta, \varphi) + k \cdot \hat{E}_c$$

$k=0.30$ decay $\times 0.95$ every 500 cycles

Exponential smoothing | decay prevents saturation

Eq 9 — Decision Policy

$$S = P + \lambda \cdot U \quad \lambda = 0.25$$
$$A_t = \operatorname{argmax}(S) \rightarrow \text{PATROL} / \text{FOCUS} / \text{ALERT} / \text{STOP}$$

Exploration (U) + exploitation (P) via λ

Physical reality gate: DANGEROUS only triggers if at least one raw sensor actually exceeds its threshold (gas>400, temp>50°C, or EMI spike >200 units above baseline) — normalisation saturation alone cannot trigger an alert.

Adaptive Scan Modes — P-Driven Physical Response

COARSE

$$P \leq 0.35$$

Poses: 3 poses

Time: ~3s

3 base × 1 shoulder × 1 elbow
No tip sweep
Low field → move fast, cover ground quickly

FOCUSED

$$0.35 < P \leq 0.65$$

Poses: 12 poses

Time: ~12s

3 base × 2 shoulder × 2 elbow
No tip sweep
Rising field → home in on source area

DENSE

$$P > 0.65$$

**Poses: 18 poses
+tip sweep**

Time: ~25s

3 base × 3 shoulder × 2 elbow
i/k tip sweep 5 angles at each base
High field → maximum directional precision

P (scan priority) increases → stronger field / rising trend / high uncertainty

Multi-Modal Sensor Fusion & Hazard Classification

Sensor Fusion — Unified Hazard Score H

$$H = w_1 \cdot \hat{E} + w_2 \cdot \hat{G} + w_3 \cdot \hat{T} + w_4 \cdot \hat{V} \quad w_1=0.35 \quad w_2=0.25 \quad w_3=0.25 \quad w_4=0.15$$

$H \in [0,1]$ — weighted combination of all normalised sensor channels

$\hat{E}$ EM Field

EMI coil A4

0–1023 ADC
Running min-max norm
Spike: >200 above baseline

$\hat{G}$ Gas Level

MQ-2 sensor

0–1023 ADC
Threshold 400 raw
Norm: /1023 clamped

$\hat{T}$ Temperature

DHT22

–40 to 80°C range
Threshold: 50°C
Running min-max

$\hat{V}$ Vision

YOLO detection

fire/smoke=1.0
person=0.3
none=0.0

Zone Classification:

SAFE $H < 0.30$

All sensors normal

WARNING $0.30 \leq H < 0.65$

Elevated readings

DANGEROUS $H \geq 0.65$ + sensor breach

Actual threshold exceeded

Eq 10 — Temporal Trend Analysis

$$d\hat{E}/dt = \text{mean}(\hat{E}_t - \hat{E}_{\{t-n\}}) \quad n \in \{1..4\} \quad | \quad \text{trend_rising if slope} > 0.003 \text{ AND regression} > 0.002$$

Windowed mean prevents single-sample spikes triggering false rising-trend alerts

Results

Software results · Hardware measurements · Sensor data

Results — Simulation & Verification

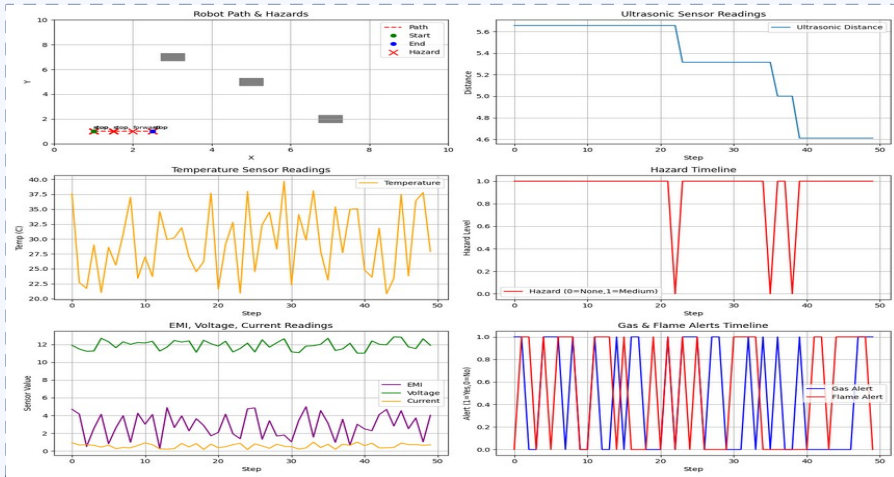


Fig. 19 — Phase 1 simulation results graph — baseline validation

```
[PATROL] + Patent arm EM scan at spot 1...
[ARM EIL SCAN] Adaptive - mode=FOCUSED P=0.423
[ARM EIL SCAN] FOCUSED: 12 poses - homing in
[F] base= 45° sh= 70° el= 70° | EMI=142.0 | θ=-12.3°
[F] base= 45° sh= 70° el=110° | EMI=187.0 | θ=-34.7°
[F] base= 45° sh=110° el= 70° | EMI=203.0 | θ=-19.2°
[F] base= 90° sh= 70° el= 70° | EMI=241.0 | θ= -8.1° -PEAK
[F] base= 90° sh= 70° el=110° | EMI=198.0 | θ=-15.4°
[F] base=135° sh= 70° el= 70° | EMI=112.0 | θ=+15.2°
[ARM EIL SCAN] Peak EMI=241.0 at base=90° sh=70°
[ARM EIL SCAN] θ=-8.1° φ=5.0° - EM source direction
Arm scan complete at spot 1. Peak at tilt -8 degrees.
```

Fig. 21 — Arm EM scan peak detection terminal output

```
gazebo -robot: /usr/share/gazebo/robot
gazebo -robot: /usr/share/gazebo/robot
gazebo -robot: /usr/share/gazebo/robot
[CPU] GPU-GS80 initialized for orientation-aware EM mapping.
[CPU] Background orientation-EM thread started.
Deleted: History/cp_05-04-2026T13:50:57_performing_routine_patrol_and_hazard_detection_tasks_in_dinly_lit_area.txt
Deleted: History/cp_05-04-2026T13:50:57_processing_visual_data_from_detected_person_below_me_slowly.txt
Deleted: History/cp_05-04-2026T13:50:57_robot_waiting_for_further_instructions_from_chris_in_lockdown_mode.txt
Deleted: History/cp_05-04-2026T13:50:57_robot_waiting_for_further_instructions_from_chris_in_lockdown_mode_currently.txt
Deleted: History/cp_05-04-2026T13:50:57_robot_waiting_for_further_instructions_from_chris_in_lockdown_mode_currently.txt
Deleted: History/cp_05-04-2026T13:50:57_robot_waiting_for_further_instructions_from_chris_in_lockdown_mode_currently.txt
Deleted: History/cp_05-04-2026T13:50:57_robot_waiting_for_further_instructions_from_chris_in_lockdown_mode_currently.txt
All txt files in the directory have been deleted.
Server IP loaded: 10.161.241.206
Connecting to Websocket server at 10.161.241.206:8040...
Websocket connected.
Initialized recognizer and VAD.
ALSA lib conf.c:5189:(snd_func_refer) Unable to find definition 'cards.0.pcm.front.0:CARD=0'
ALSA lib conf.c:5189:(snd_func_refer) function snd_func_refer returned error: No such file or directory
ALSA lib pcm.c:2666:(snd_pcm_open_noupdate) Unknown PCM front
ALSA lib pcm.c:2666:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.rear
ALSA lib pcm.c:2666:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.center_lfe
ALSA lib pcm.c:2666:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.side
ALSA lib conf.c:5189:(snd_func_refer) Unable to find definition 'cards.0.pcm.surround51.0:CARD=0'
ALSA lib conf.c:5189:(snd_func_refer) function snd_func_refer returned error: No such file or directory
ALSA lib pcm.c:2666:(snd_pcm_open_noupdate) Unknown PCM surround21
ALSA lib conf.c:5189:(snd_func_refer) Unable to find definition 'cards.0.pcm.surround51.0:CARD=0'
ALSA lib conf.c:5189:(snd_func_refer) function snd_func_refer returned error: No such file or directory
ALSA lib pcm.c:2666:(snd_pcm_open_noupdate) Unknown PCM surround21
ALSA lib conf.c:5189:(snd_func_refer) Unable to find definition 'cards.0.pcm.surround40.0:CARD=0'
ALSA lib conf.c:5189:(snd_func_refer) function snd_func_refer returned error: No such file or directory
ALSA lib pcm.c:2666:(snd_pcm_open_noupdate) Unknown PCM surround40
ALSA lib conf.c:5189:(snd_func_refer) Unable to find definition 'cards.0.pcm.surround51.0:CARD=0'
ALSA lib conf.c:5189:(snd_func_refer) function snd_func_refer returned error: No such file or directory
ALSA lib conf.c:5189:(snd_func_refer) function snd_func_refer returned error: No such file or directory
```

Fig. 20 — Robot program starting... view

```
Image sent successfully in 0.003 seconds.
false
false
false
true -- Consolidated rough draft memory from past sessions.
name flag: false
name text: false
Image sent successfully in 0.003 seconds.
=====
[PATROL] spot=1 sector=1/8 zone=spot1_sector0 cam=4
Temp : 30.6°C ok
Humidity : 76.0% ok
Gas : 245 ok
EM : 566.0 ok Distance: 200cm
EIL zone : WARNING P=0.554 FOCUS_SCAN
M_t grid : 0.9842 zone_visits=11
Image sent successfully in 0.003 seconds.
=====
false
false
false
true -- Consolidated rough draft memory from past sessions.
name flag: false
name text: false
Image sent successfully in 0.003 seconds.
[AUDIO] Google: could not understand audio
```

Fig. 22 - sensor readings in running program

Software Result — Live Arm EM Scan Terminal Output

```
[PATROL] ↑ Patent arm EM scan at spot 1...
```

```
[ARM EIL SCAN] Adaptive — mode=FOCUSED P=0.423
```

```
[ARM EIL SCAN] FOCUSED: 12 poses — homing in
```

```
[F] base= 45° sh= 70° el= 70° | EMI=142.0 | θ=-12.3°
```

```
[F] base= 45° sh= 70° el=110° | EMI=187.0 | θ=-34.7°
```

```
[F] base= 45° sh=110° el= 70° | EMI=203.0 | θ=-19.2°
```

```
[F] base= 90° sh= 70° el= 70° | EMI=241.0 | θ= -8.1° ←PEAK
```

```
[F] base= 90° sh= 70° el=110° | EMI=198.0 | θ=-15.4°
```

```
[F] base=135° sh= 70° el= 70° | EMI=112.0 | θ=+15.2°
```

```
[ARM EIL SCAN] Peak EMI=241.0 at base=90° sh=70°
```

```
[ARM EIL SCAN] θ*=-8.1° φ*=5.0° → EM source direction
```

```
Arm scan complete at spot 1. Peak at tilt -8 degrees.
```

Output explained:

[F]	FOCUSED mode (P=0.423 → 12 poses)
base/sh/el	Arm joint angles — unique 3D orientation
EMI=241	Raw ADC from coil at that arm pose
θ=-8.1°	MPU pitch — actual arm orientation
← PEAK	Highest EMI found = EM source direction
θ*=-8.1°	Source at -8° pitch from arm origin

Software Result — Patrol Sensor Display (Live Terminal)

```
=====
[PATROL] spot=3 sector=5/8 zone=spot3_sector4 cam=2
```

```
Temp      : 30.8°C   ok (threshold 50.0°C)
```

```
Humidity  : 81.6%   ok (threshold 90.0%)
```

```
Gas       : 228     ok (threshold 400)
```

```
EM coil   : 241.0   ok Distance: 45cm
```

```
EIL zone  : WARNING P=0.423 FOCUSED FOCUS_SCAN
```

```
M_t grid  : 0.4231  zone_visits=3
```

```
=====
[ARM EIL SCAN] Starting... FOCUSED mode
```

```
[F] base= 90° sh=70° el=70° | EMI=241.0 | θ=-8.1°
```

```
[ARM EIL SCAN] θ*=-8.1° φ*=5.0° → source identified
```

```
Arm scan complete at spot 3. WARNING zone.
```

Display fields:

spot=3	3rd forward patrol position
sector=5/8	5th of 8 × 45° scan sectors
zone=spot3_sector4	Unique zone ID for memory
cam=2	Camera tilt-up (cmd 2)
P=0.423	Priority → FOCUSED mode
M_t=0.4231	Hazard grid value [0-1]
zone_visits=3	Visited 3 times — U=0.25

Results — Data Files & Heatmaps

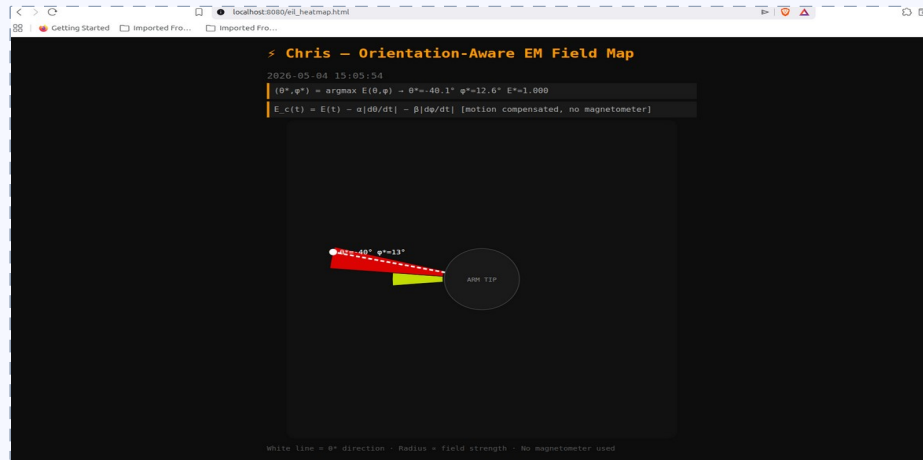


Fig. 23 — EIL heatmap HTML — polar EM orientation map

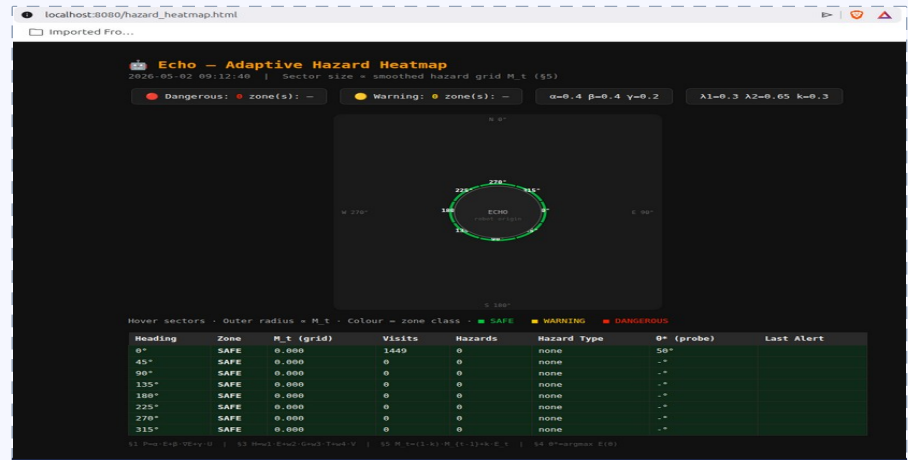


Fig. 24 — Hazard heatmap HTML — zone memory polar map

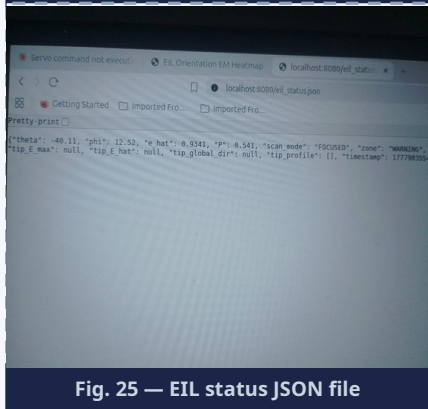


Fig. 25 — EIL status JSON file

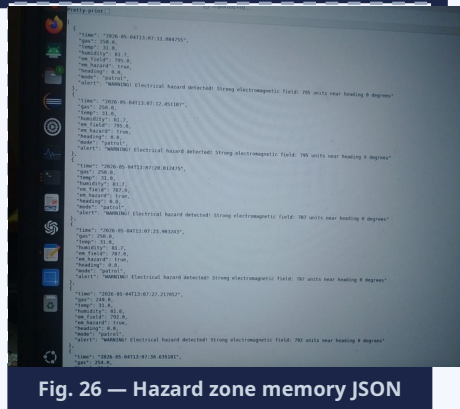


Fig. 26 — Hazard zone memory JSON

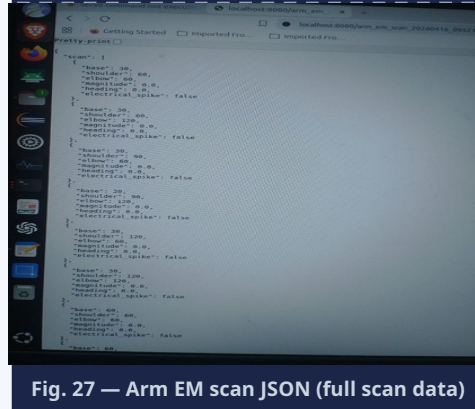


Fig. 27 — Arm EM scan JSON (full scan data)

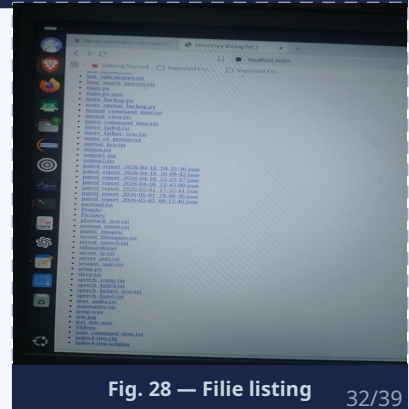


Fig. 28 — File listing

Results — Hardware Testing & Measurement

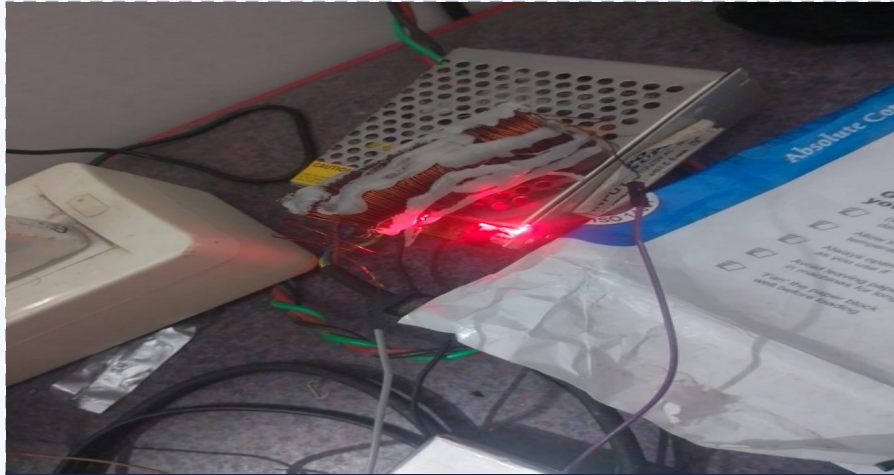


Fig. 29 — EMI coil measuring EM field near 12V adapter

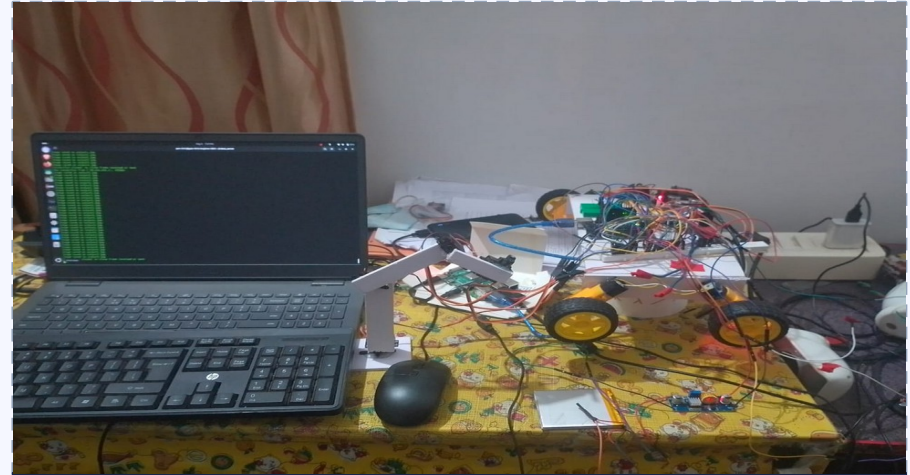


Fig. 30 — Full robot with laptop running patrol demonstration



Fig. 31 — Hardware circuit — LM358 EMI board assembled

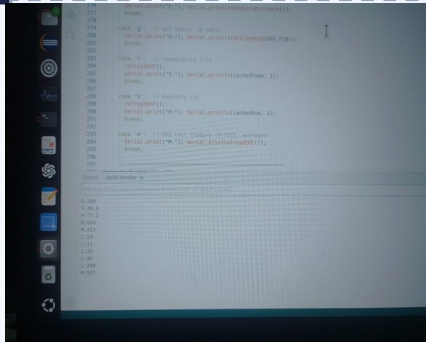


Fig. 32 — Arduino IDE: serial sensor commands output



Fig. 33 — Robotic arm (close-up)

Output Files Generated by the Robot

<code>eil_heatmap.html</code>	Polar orientation-space EM map	<i>Every 200 EIL cycles (~20s)</i>	SSH port-forward → localhost:8080
<code>hazard_heatmap.html</code>	Zone memory polar hazard map	<i>End of every patrol</i>	Browser via http.server 8080
<code>eil_status.json</code>	EIL zone, P, θ^* , φ^* , E* realtime	<i>Every 200 cycles</i>	<code>cat eil_status.json</code>
<code>hazard_zone_memory.json</code>	Zone visits, hazard counts, priority	<i>Every patrol step</i>	Persists across power cycles
<code>arm_em_scan_DATE.json</code>	Full 18-pose scan data with θ, φ, E	<i>After each arm scan</i>	<code>cat arm_em_scan_*.json</code>
<code>patrol_report_DATE.json</code>	Full sensor log per spot and sector	<i>End of patrol</i>	SCP to laptop
<code>emi_latest.txt</code>	Latest EMI coil ADC value	<i>Every sensor read cycle</i>	<code>watch cat emi_latest.txt</code>
<code>output1.jpg</code>	Latest annotated camera frame	<i>Continuously</i>	<code>http://PI_IP:8040</code>

AI Integration & Voice Command System

Groq Cloud AI — LLaMA 70B

- Receives: sensor context + camera image + mode + last phrase
- Returns JSON: physical_cmd | speech_cmd | mental_cmds
- Hardware cmds: Move Fwd | Camera Pan | Arm EM Scan | Turn...
- Triggers: voice / hazard change / YOLO person / heartbeat
- Speech dedup: 60s cooldown per text string
- Patrol summaries excluded from voice-cmd trigger path

Voice Command Shortcuts

"chris patrol"	Start autonomous patrol
"camera right/left/center"	Pan 5/3/4
"camera up/down"	Tilt 2/1
"introduce yourself"	14-part hardware intro
"introduce equations"	16-part algorithm explanation
"chris stop"	Stop all movement
"arm scan"	Trigger full arm EM scan
"move forward/backward"	Drive motors w/s
"turn left/right"	Turn 15° or 45°

Conclusion



A fully operational AI hazard detection robot has been built, integrated and tested end-to-end.



Orientation-aware EM field mapping using MPU-6500 on robotic arm successfully identifies EM source direction without any magnetometer — a novel approach.



Adaptive scanning (COARSE / FOCUSED / DENSE driven by P score) dynamically adjusts scan density based on real-time signal strength and uncertainty.



Multi-modal sensor fusion (EM, gas, temperature, humidity, vision) with compound hazard classification provides reliable hazard detection.



Autonomous patrol, directional EM heatmaps, voice interaction via Groq LLaMA AI, and zone memory make CHRIS a complete hazard monitoring platform.



All 10 mathematical equations — complementary filter, motion compensation, normalisation, scan priority, hazard grid, decision policy — are implemented and running live.

Future Work



Real Compass Integration

Add QMC5883L alongside MPU-6500 for absolute heading — enabling GPS-referenced hazard mapping across sessions.



SLAM-Based Navigation

Replace step-counter zone memory with ROS2 + LiDAR SLAM for true positional awareness and revisit detection.



Wireless EMI Calibration

Auto-calibrate EMI baseline by running 10 full-room sweeps and computing environment-specific adaptive thresholds.



Battery Monitoring

Integrate INA219 current sensor to monitor real-time battery level and trigger auto-dock when below 20%.



Cloud Hazard Database

Upload patrol reports and EM scans to cloud; enable multi-robot shared hazard maps for large facility coverage.



Industrial & Disaster Deployment

Deploy in industrial substations, manufacturing plants and disaster response zones for real-world algorithm validation.

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Thank You

Questions & Discussion

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Robot: CHRIS | Groq LLaMA 70B AI | ~7000 lines Python | Raspberry Pi 4B